DECK 08 · WEEK 4 · MORNING 1

# FSA III — Taxes, Liabilities & Quality

*The analyst toolkit that turns a set of statements into a view on the business*

CFA Level I · Vol 4 Learning Modules 8–12

# Roadmap — FSA III — Taxes, Liabilities, Quality, Ratios & Modeling

*6 sections · ~2 hours · Built bottom-up from CFA readings*

| | |
|---|---|
| 01 | **Long-term liabilities & leases**<br>IFRS 16 / ASC 842 · ROU asset + lease liability · finance vs operating P&L shape · lessor accounting |
| 02 | **Pensions & share-based comp**<br>DC vs DB · PBO roll-forward · service / interest / actuarial · stock grants and options at grant-date FV |
| 03 | **Income taxes**<br>Accounting profit vs taxable income · DTL / DTA · Reston Partners depreciation · effective-tax-rate walk |
| 04 | **Reporting quality & red flags**<br>Quality spectrum · aggressive vs conservative · fraud triangle · accruals ratio · CFO / NI signal |
| 05 | **Ratio analysis & DuPont**<br>Activity · liquidity · solvency · profitability · 3-factor and 5-factor DuPont (Amsterdam PLC) |
| 06 | **3-statement modeling**<br>Build sequence · working-capital drivers · CCC · base / bull / bear revenue scenarios · cash tie-out |

# Learning objectives

*By the end of this deck you should be able to:*

1. Capitalise a lease under IFRS 16 and ASC 842 — compute the ROU asset, lease liability, and the P&L and cash-flow impacts under finance vs operating classification.

2. Distinguish defined-contribution from defined-benefit pensions, roll forward the defined-benefit obligation from its five sources, and read a share-based-compensation footnote.

3. Trace a deferred tax liability from accelerated tax depreciation, identify permanent vs temporary differences, and decompose an effective-tax-rate reconciliation.

4. Place a set of financial statements on the 5-tier reporting-quality spectrum and apply the accruals ratio and CFO-to-net-income to flag aggressive accounting.

5. Compute the full suite of activity / liquidity / solvency / profitability ratios and run a 3-factor and 5-factor DuPont decomposition of ROE.

6. Build the skeleton of a 3-statement forecast model, link working-capital ratios to operating cash flow, and run a revenue-growth scenario analysis.

SECTION 01

# Long-term liabilities & leases

# Long-term liabilities come in four flavours — bonds, leases, pensions, and deferred tax — and each must be translated into comparable economic debt

*Reported "total debt" rarely equals economic debt. A careful analyst reaches for the footnotes and normalises lease liabilities, unfunded pensions, and deferred tax onto a single yardstick.*

| Instrument | Where it sits | Cash nature | Analyst adjustment |
|---|---|---|---|
| Bonds payable | Balance sheet at amortised cost | Coupons + principal | Mark-to-market if yields have moved materially |
| Leases | ROU asset + lease liability | Fixed monthly payments | Include in debt; capitalise implicit-interest portion |
| Pensions (DB) | Net funded status (A or L) | Future pension payouts | Treat underfunded amount as quasi-debt in EV |

**THE ECONOMIC-DEBT LENS**

Economic debt = bonds + capitalised leases + underfunded pension − surplus cash. Reported debt (bonds only) can understate true leverage by 40%+ for airlines, retailers, and REIT tenants where operating leases used to sit off-balance-sheet before IFRS 16.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Introduction", pp. 246–247.*

# Under IFRS 16 and ASC 842 every lease above 12 months lands on the balance sheet as a right-of-use asset and a lease liability equal to the PV of payments

*Short leases (<12 months) and low-value assets (<USD 5,000, IFRS only) can still be expensed straight-line — everything else capitalises.*

$$\text{ROU asset} = \text{Lease liability} = \Sigma \ \text{Lease_Payment_t} / (1 + r)^t$$

*where: r = rate implicit in the lease, or the lessee's incremental secured borrowing rate. t = 1 … lease term.*

### FIVE FINANCE-LEASE CRITERIA (any one triggers)

1. Lease transfers title to lessee at end of term
2. Bargain purchase option the lessee is likely to exercise
3. Lease term covers a major part of asset's economic life
4. PV of payments ≥ substantially all of asset's fair value
5. Asset is specialised — no alternative use to lessor

### IFRS 16 vs US GAAP IN ONE LINE

IFRS 16 uses a SINGLE model for lessees — every lease is treated like a finance lease on the P&L (separate interest + amortisation). US GAAP (ASC 842) preserves an "operating lease" category for lessees, where the lessee reports a single flat "lease expense" line and a flat cash outflow in CFO. Balance sheet impact is identical under both standards.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Lease Classification", pp. 249–251.*

# Under US GAAP a finance lease front-loads total expense while an operating lease delivers a flat expense — same cash, different P&L shape

*Lease: 5y, EUR 100k annual payment, 10% implicit rate → ROU asset = lease liability = EUR 379,079 at inception under both classifications.*

| Year | Finance — interest | Finance — amort. | Finance — total | Operating — flat expense | Gap (F − O) |
|---|---|---|---|---|---|
| 1 | 37,908 | 75,816 | 113,724 | 100,000 | +13,724 |
| 2 | 31,699 | 75,816 | 107,515 | 100,000 | +7,515 |
| 3 | 24,869 | 75,816 | 100,685 | 100,000 | +685 |
| 4 | 17,355 | 75,816 | 93,171 | 100,000 | −6,829 |
| 5 | 9,091 | 75,816 | 84,907 | 100,000 | −15,093 |
| **Total over 5y** | **120,921** | **379,079** | **500,000** | **500,000** | **0** |

**KEY INSIGHT — SAME CASH, DIFFERENT P&L**

Total cash outflow is EUR 500k in both classifications. Total expense is EUR 500k. But year-1 P&L is EUR 13,724 worse under the finance lease, because interest declines faster than straight-line amortisation.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, Examples 2 & 3 (Proton Enterprises), pp. 251–255.*

# Operating vs finance leases push EBITDA, asset turnover, leverage, and cash flow in opposite directions — analysts adjust before comparing peers

*The choice is a classification decision, not an economic one — but the reported ratios move enough to mislead a naive comparison.*

| Ratio | Formula | Finance lease | Operating lease | Why it differs |
|---|---|---|---|---|
| EBITDA margin | EBITDA / Revenue | Higher | Lower | Finance: lease = interest + amort (below EBITDA). Operating: lease is op-ex (above EBITDA) |
| Operating margin | Operating income / Revenue | Higher | Lower | Same reason as EBITDA |
| | | | | Finance-lease ROU amortises |

**THE ANALYST ADJUSTMENT**

To compare an IFRS-reporting airline (everything looks finance-lease-ish) to a US-GAAP peer with operating-lease classification, strip lease expense out of op-ex on the GAAP firm and re-class as D&A + interest. Put both on the same basis before computing multiples.

| | | | | |
|---|---|---|---|---|
| Debt / Equity | Lease liability counted as debt | Higher | Higher | balance sheet (identical BS effect) |
| | | | | Finance: only |

# Lessor accounting is nearly identical under IFRS and US GAAP — finance leases derecognise the asset, operating leases keep it on-book

*The analyst's question: is this lessor running a financing business (banks, aircraft-lessors) or a rental business (real-estate REITs)?*

| Financial statement | Finance lease (lessor) | Operating lease (lessor) |
|---|---|---|
| Balance sheet | De-recognise asset; recognise lease receivable at PV | Keep asset; continue depreciating it |
| Income statement | Interest income (as effective rate on receivable) | Straight-line lease revenue + depreciation expense |
| Gain / loss at inception | Recognise sale-type gain/loss (asset FV vs carrying) | None — transaction is not a sale |
| Cash flow statement | Full cash receipt in CFO | Full cash receipt in CFO |
| Revenue vs interest disclosure | If leasing is primary business → shown as revenue | Revenue line always |

**ILLUSTRATION — AERCAP (AIRCRAFT LESSOR)**

AerCap owns ~1,500 aircraft leased to airlines. Under finance-lease classification (e.g. a 12-year lease on a 15-year-life A321), AerCap derecognises the aircraft and books a lease receivable. Under operating-lease classification (short operating-lease of ageing equipment) it keeps the plane on the balance sheet and books rental revenue.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Lessor Accounting", pp. 254–256.*

SECTION 02

# Pensions & share-based comp

# Defined-contribution plans shift investment and longevity risk to employees; defined-benefit plans keep that risk on the employer's balance sheet

*Over the last 30 years DC plans have replaced DB plans almost universally in the private sector precisely because DC moves balance-sheet risk off the sponsor.*

| Dimension | Defined contribution (DC) | Defined benefit (DB) |
|---|---|---|
| Who promises what | Employer promises a fixed ANNUAL CONTRIBUTION to plan | Employer promises a fixed ANNUAL PENSION in retirement |
| Who bears investment risk | Employee — retirement income depends on plan performance | Employer — must top up shortfalls out of corporate cash |
| Who bears longevity risk | Employee (and annuity provider if annuitised) | Employer — pays until the last retiree dies |
| P&L expense | Cash contribution is pension expense, full stop | Service cost + net interest + past-service cost |
| Balance-sheet line | None beyond accrued payable | Net funded status (PBO – plan assets) as A or L |
| | | Extensive footnote — assumptions |

**ANALYST TELL**

If the pension footnote is longer than one page, the firm sponsors a DB plan — and likely inherits a structurally unfunded liability from a pre-2000 workforce.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Financial Reporting for Postemployment Plans", pp. 262–266.*

# A defined-benefit pension obligation rolls forward each year from five sources — service cost, interest, past-service, actuarial gains/losses, and benefits paid

*The PBO roll-forward is the single most important pension footnote — it tells the analyst what's service cost (recurring) vs actuarial remeasurement (volatile, OCI).*

**HELMSAG · DB PENSION ROLL-FORWARD · YEAR ENDED 31 DEC**

Opening PBO: EUR 800m. Discount rate 4%. Service cost 35m. 20m of benefits paid. Actuarial loss 15m from lower discount rate. Past-service cost 5m from plan amendment.

**1.** Opening PBO (Jan 1): +800.0
**2.** Service cost (present value of benefits earned this year): +35.0
**3.** Interest cost (opening PBO × discount rate = 800 × 4.0%): +32.0
**4.** Past-service cost (benefit-formula enhancement): +5.0
**5.** Actuarial loss (change in discount rate / longevity): +15.0
**6.** Benefits paid to retirees: −32.0
**7.** Closing PBO (Dec 31) = 800 + 35 + 32 + 5 + 15 − 32 → +855.0

**Closing PBO = EUR 855m. Service cost (35) + interest (32) + past-service (5) hit P&L; actuarial loss (15) hits OCI under IFRS.**

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Financial Reporting for DB Pension Plans", pp. 266–271.*

# Share-based compensation is measured at grant-date fair value and expensed over the vesting period — no cash, but a real hit to earnings and dilution

*SBC is the textbook example of a non-cash expense that is NOT a cash-flow adjustment to value — it transfers wealth from existing shareholders to employees.*

## STOCK GRANTS / RSUs

Fair value at grant = share price × shares granted. Expensed straight-line over vesting period. Cliff or graded vesting. No option-pricing model needed.

## STOCK OPTIONS

Fair value at grant requires an option-pricing model (Black-Scholes or binomial). Inputs: share price, strike, volatility, life, risk-free rate, dividend yield, expected forfeitures.

## INDUSLTD · 100,000 RSUs · 3-YEAR GRADED VESTING · SHARE AT USD 45

CFO is granted 100,000 RSUs vesting 1/3 per year at grant-date share price of USD 45.

**1.** Grant-date fair value = 100,000 × USD 45 = USD 4.5m
**2.** Year 1 expense = 4.5m × (1/3) = USD 1.50m → recognised as SG&A with offset to APIC

**Cumulative SBC expense USD 4.5m. Zero cash outflow. Dilution = 100,000 new shares issued at vesting.**

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Share-Based Compensation", pp. 272–278.*

SECTION 03

# Income taxes

# Accounting profit and taxable income diverge because accounting standards and tax laws allow different timing for the same economic transaction

*Accounting profit follows IFRS/US GAAP; taxable income follows the local tax code. Differences are either timing (temporary) or structural (permanent).*

$$\text{Income tax expense} = \text{Taxes payable (current)} + \Delta\ \text{Deferred tax liabilities} - \Delta\ \text{Deferred tax assets}$$

*where: Taxes payable follows taxable income × statutory rate. The deferred-tax deltas reconcile accounting expense back to book profit × tax rate.*

| Reconciling item | Type | Direction | Example |
|---|---|---|---|
| Accelerated tax depreciation | Temporary | DTL | Tax: 5y MACRS. Book: 10y straight-line |
| Provision for doubtful debts (IFRS only when paid) | Temporary | DTA | Expense booked now, tax-deductible only on write-off |
| Tax-loss carryforwards | Temporary | DTA | Prior losses usable against future taxable income |
| Municipal-bond coupon income | Permanent | Lower ETR | Non-taxable under US tax code |
| Fines, penalties, non-deductible entertainment | Permanent | Higher ETR | Book expense but never tax-deductible |

# Temporary differences generate deferred tax assets or liabilities; permanent differences only move the effective tax rate

*The rule of thumb: compare carrying amount to tax base. An asset worth more on the books than on the tax return means future tax is owed — a DTL.*

| Balance-sheet item | Carrying vs tax base | Gives rise to | Economic intuition |
|---|---|---|---|
| Asset | Carrying > tax base | Deferred tax LIABILITY | You'll recover more book value than tax basis → future taxable income |
| Asset | Carrying < tax base | Deferred tax ASSET | You've paid tax on more than book value → future tax relief |
| Liability | Carrying > tax base | Deferred tax ASSET | You've expensed more than tax-deductible → future deduction available |

**VALUATION ALLOWANCE (US GAAP) / DTA DE-RECOGNITION (IFRS)**

A DTA is only useful if the firm will have taxable income against which to use it. If realisation is NOT more-likely-than-not, US GAAP establishes a valuation allowance that writes the DTA down; IFRS simply does not recognise it. Both treatments land at the same economic reality — a DTA whose realisation is in doubt is worth zero. A reversal of a valuation allowance (firm turns profitable) is a one-time EPS boost that analysts strip out.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 9, Exhibit 1, p. 285.*

# Accelerated tax depreciation is the canonical DTL: tax base falls faster than book value, creating a growing deferred tax liability until the asset fully depreciates

*Same economic asset, different schedules: book straight-line 10y, tax straight-line 7y. The tax authority gives you a cash advantage today and claws it back later.*

| Item (GBP m) | Year 1 | Year 2 | Year 3 |
|---|---|---|---|
| Carrying amount (10-yr book, 2,000/yr) | 18,000 | 16,000 | 14,000 |
| Tax base (7-yr tax, 2,857/yr) | 17,143 | 14,286 | 11,429 |
| Temporary difference (carrying – tax base) | 857 | 1,714 | 2,571 |
| Deferred tax liability @ 30% tax rate | 257 | 514 | 771 |
| Change in DTL (added to tax expense) | +257 | +257 | +257 |

**READING THE DTL IN YEAR 3**

Pretax book income 26,700. Taxable income 25,843 (because tax depreciation is 2,857 vs book 2,000). Statutory rate 30%.

**Effective book rate reconciles back to 30% once the DTL movement is included.**

**3.** Total income-tax expense on IS = 7,753 + 257 = 8,010 (= 26,700 × 30%)

*Source: CFA Institute, Curriculum Vol 4, Learning Module 9, Example 3 (Reston Partners), pp. 288–291.*

# Every effective-tax-rate reconciliation walks from statutory to reported line by line — analysts use it to sanity-check permanent items and jurisdiction mix

*NortCorp — hypothetical US-domiciled industrial with 40% of profit in Ireland (12.5%) and 20% in the UK (25%) — reconciling 21% statutory → 21.3% effective.*

| Reconciling item | Δ ETR (pp) | USD m (on 1,000 pretax) | Type |
|---|---|---|---|
| US statutory federal rate | 21.00 | 210.0 | Base |
| Foreign rate differential (Ireland at 12.5%, UK at 25%) | −2.50 | −25.0 | Perm |
| State and local taxes (net of federal benefit) | +3.20 | +32.0 | Perm |
| Non-deductible expenses (M&E, lobbying, fines) | +1.10 | +11.0 | Perm |
| Research & development tax credits | −1.80 | −18.0 | Perm |
| Change in uncertain tax positions (GILTI, BEAT) | +0.40 | +4.0 | Perm |
| Valuation-allowance release (loss carryforwards now useable) | −0.10 | −1.0 | Disc |
| **Reported effective tax rate** | **21.30%** | **213.0** | |

**ANALYST FLAG**

Any single reconciling line > 2% of pretax profit deserves a follow-up question — especially one-off items that could reverse next year.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 9, "Tax Rate Reconciliation", pp. 293–298.*

SECTION 04

# Reporting quality & red flags

# Financial-reporting quality and earnings quality are two separate axes — the worst companies score poorly on both simultaneously

*Reporting quality = are the numbers faithful? Earnings quality = are the economics sustainable? A firm can be honest about bad economics (top-left) — that is still a high-quality report.*

| | High earnings quality | Low earnings quality |
|---|---|---|
| High reporting quality | ✔ Ideal. Reports are faithful AND economics are sustainable. Valuation anchors here. | Reports faithful but economics weak — analysts can still value the firm, just at a lower multiple. |
| Low reporting quality | IMPOSSIBLE — if reports are unfaithful you cannot conclude earnings quality is high. | ✘ Worst. Misleading numbers AND weak economics. Often precedes restatements and fraud. |

## WHY THE TWO AXES MATTER

The top-left quadrant (high reporting, low earnings) is perfectly legitimate — e.g. Toyota's FY14 report where a weak yen drove an 88% operating-profit increase was high-quality reporting of low-quality (unsustainable FX-driven) earnings. The bottom-right quadrant is where fraud lives — Enron, WorldCom, Wirecard. The analyst's job is to (1) establish reporting quality FIRST, then (2) form a view on earnings quality.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 10, Exhibit 1, pp. 319–320.*

# Reporting quality lives on a continuum from GAAP-faithful through biased choices to earnings management and finally fictitious transactions

*The jump from tier 3 to tier 4 is a jump across a legal boundary — earnings management within GAAP becomes misstatement outside it.*

| | |
|---|---|
| 1 | **Tier 1 — GAAP, decision-useful, sustainable**<br>Top of the spectrum. Reports conform to GAAP, provide decision-useful info, AND earnings are sustainable and deliver adequate returns on capital. |
| 2 | **Tier 2 — GAAP, decision-useful, but sustainable?**<br>Reports are fine; economics are not. E.g. reported earnings came from one-off FX, asset sales, or non-recurring items. Low earnings quality, not a reporting issue. |
| 3 | **Tier 3 — Within GAAP, but biased choices**<br>Company picks accounting options (LIFO vs FIFO, capitalise vs expense, depreciable life) to flatter reported earnings. Includes earnings smoothing via reserves. |
| 4 | **Tier 4 — Within GAAP, but earnings management**<br>Real earnings management (cutting R&D to hit a number) and accounting earnings management (aggressive estimates). Still technically GAAP-compliant but purposeful. |
| 5 | **Tier 5 — Non-compliant / fictitious**<br>Outright violation of GAAP. Fake revenue, hidden liabilities, off-balance-sheet SPVs used to conceal debt. Enron, WorldCom, Wirecard, Satyam. |

*Source: CFA Institute, Curriculum Vol 4, Learning Module 10, Exhibit 2 (Quality Spectrum), p. 321.*

# The fraud triangle — motivation, opportunity, rationalisation — explains why managers in legitimate companies issue low-quality reports

*All three corners must be present for financial fraud to occur. Kill any one of them and the fraud doesn't start.*

## 1 · MOTIVATION

Pressure to hit a number:

- Debt covenant close to trigger
- Earnings-based exec compensation
- Analyst consensus to beat
- Stock-price-linked wealth
- Regulatory capital threshold

Remove motivation and fraud disappears — which is why long vesting periods for exec comp reduce fraud risk.

## 2 · OPPORTUNITY

Ability to perpetrate without detection:

- Weak internal controls
- Ineffective, captive board
- Auditor tenure > 10 yrs
- Accounting judgement-heavy industry (software, insurance)
- Complex group structure
- Limited analyst coverage

Mechanisms that reduce opportunity: independent audit, PCAOB oversight, SOX 302/404, whistleblower hotlines.

## 3 · RATIONALISATION

Moral justification to the self:

- "Everyone in the industry does it"
- "Bridge year — we'll true it up later"
- "The accounting rules are unfair"
- "Protecting jobs is the right thing"
- "We deserved that bonus"

Cultural tell: management openly endorses "within-limits" discretion in reporting — that is the tone-at-the-top that enables fraud.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 10, "Motivations and Conditions", pp. 330–340.*

# The accruals ratio and CFO-to-net-income are the two simplest earnings-quality flags — both cheap to compute and alarmingly predictive

*Sloan (1996) showed firms with the highest accruals ratios under-performed the market by ~10% in the following year. The intuition: accruals that aren't backed by cash eventually reverse.*

**Accruals ratio (BS) = (NOA_end − NOA_begin) / ½(NOA_end + NOA_begin)**
**CFO / NI → should cluster near 1.0**

*where: NOA = net operating assets = (total assets − cash) − (total liabilities − total debt). An accruals ratio > 10% is a red flag.*

## VENTARA INC. · FY2024 EARNINGS-QUALITY CHECK

NOA_begin = USD 800m; NOA_end = USD 1,040m. Net income 180m. CFO reported at 95m.

**1.** Avg NOA = ½(800 + 1,040) = 920m
**2.** Accruals ratio = (1,040 − 800) / 920 = 240 / 920 = +26.1% → well above 10% red-flag threshold
**3.** CFO / NI = 95 / 180 = 0.53 → far below 1.0; accrual-heavy earnings
**4.** Cross-check: DSO up from 58 → 78 days (+34%) — classic channel-stuffing pattern

**Ventara fails BOTH tests. The 26% accruals ratio and 0.53 CFO/NI together warrant a detailed AR and revenue-recognition review before any long position.**

*Source: CFA Institute, Curriculum Vol 4, Learning Module 10, "Pay Attention to the Relationship between Cash Flow and Net Income", pp. 374–378.*

SECTION 05

# Ratio analysis & DuPont

# Ratio analysis starts with four families — activity, liquidity, solvency, profitability — each answering a different question about the firm

*Never report a ratio in isolation. Every number needs (a) a history, (b) a peer comparable, and (c) an economic interpretation.*

| Family | Question | Three canonical ratios | Typical benchmark |
|---|---|---|---|
| Activity | Am I using assets efficiently? | Inventory turnover · DSO · Asset turnover | Peers |
| Liquidity | Can I pay my bills in 12 months? | Current ratio · Quick ratio · Cash ratio | >1.0 |
| Solvency | Can I survive the decade? | Debt / Equity · Debt / Assets · Interest coverage (EBIT/I) | <2.0 / >3× |
| [illegible] | [illegible] | [illegible] | [illegible] |

**RATIO RULES OF THUMB**

1. Always use AVERAGES of balance-sheet items (opening + closing / 2) when combining with a flow item (revenue, NI).
2. Beware quarter-end window dressing: balance-sheet date figures can be unrepresentative.
3. Normalise for leases, SBC, and restructurings before cross-comparing peers.
4. A ratio only means something against a benchmark (history, peer group, or policy).

*Source: CFA Institute, Curriculum Vol 4, Learning Module 11, "Ratio Analysis", pp. 400–432.*

# Liquidity ratios measure whether a firm can survive the next twelve months; solvency ratios whether it survives the next decade

*IndusLtd (healthy industrial) vs StressCo (over-levered peer with covenant headroom collapsing). Same industry, same size, very different shape.*

| Ratio | Formula | IndusLtd | StressCo | Threshold |
|---|---|---|---|---|
| Current ratio | (Current assets) / (Current liabilities) | 2.4 | 0.9 | >1.0 |
| Quick ratio | (Cash + AR + marketable securities) / CL | 1.6 | 0.5 | >1.0 |
| Cash ratio | Cash / Current liabilities | 0.8 | 0.1 | >0.2 |
| Debt / Equity | Total debt / Total equity | 0.7 | 3.4 | <2.0 |
| Debt / Assets | Total debt / Total assets | 0.30 | 0.65 | <0.50 |
| Interest coverage | EBIT / Interest expense | 8.2 | 1.6 | >3.0 |
| Fixed-charge coverage | (EBIT + Lease exp) / (Interest + Lease exp) | 5.9 | 1.3 | >2.0 |

**READING**

StressCo fails every threshold. Even if it survives FY, the D/E of 3.4× and coverage of 1.6× leave almost no headroom for a margin slip.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 11, "Liquidity Ratios" & "Solvency Ratios", pp. 414–424.*

# Profitability ratios decompose into margin × turnover — the first question an analyst asks after computing ROA

*Two firms with identical 8% ROA can look very different: a grocer earns it via 25× turnover × 0.32% margin; a luxury-goods house via 0.8× turnover × 10% margin.*

$$\text{ROA} = \text{Net profit margin} \times \text{Total asset turnover} \qquad \text{ROE} = \text{ROA} \times \text{Leverage}$$

*where: Margin × turnover = ROA captures operating efficiency; layering leverage gets you to ROE (equity return).*

| Firm | Revenue | Net income | Avg assets | NP margin | Asset turnover | ROA |
|---|---|---|---|---|---|---|
| Grocer (MassMart) | 80,000 | 240 | 10,000 | 0.30% | 8.00× | 2.4% |
| Software (CloudWare) | 10,000 | 2,000 | 8,000 | 20.00% | 1.25× | 25.0% |
| Luxury brand (MaisonM) | 6,000 | 900 | 7,500 | 15.00% | 0.80× | 12.0% |
| Industrial (IndusLtd) | 12,000 | 720 | 9,000 | 6.00% | 1.33× | 8.0% |

**STRATEGY**

Compare same-industry ROA. Different industries have structurally different margin-turnover curves — mixing them is a classic analyst mistake.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 11, "Profitability Ratios", pp. 425–432.*

# 3-factor DuPont answers the question "why is ROE what it is?" — the three levers are margin, efficiency, and leverage

*The three-factor version is the fastest way to diagnose whether an ROE change is operating (margin, turnover) or financial (leverage).*

**ROE = (NI / Revenue) × (Revenue / Avg assets) × (Avg assets / Avg equity)**

*where: Net profit margin × Total asset turnover × Leverage = Return on equity.*

| Year | Net margin | Asset turnover | Leverage | ROE |
|---|---|---|---|---|
| FY2 | −0.47% | 0.84× | 1.60× | −0.62% |
| FY3 | 1.13% | 0.93× | 1.54× | 1.62% |
| FY4 | 1.11% | 0.95× | 1.58× | 1.66% |
| FY5 | 3.33% | 1.11× | 1.60× | 5.92% |
| **Δ FY2 → FY5** | **+380 bps** | **+0.27×** | **flat** | **+654 bps** |

**NortCorp READING**

The ROE improvement came from operating drivers (margin and turnover) — leverage was flat. A genuinely improved business rather than a re-levering story.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 11, DuPont Analysis, pp. 433–438.*

# 5-factor DuPont separates taxes and interest from operations — the version Bloomberg ships and the one IB analysts use on earnings calls

*Use when the question is "is this ROE move operating, financing, or tax?" — and it almost always is.*

$$ROE = (NI/EBT) \times (EBT/EBIT) \times (EBIT/Rev) \times (Rev/Assets) \times (Assets/Equity)$$

*where: Tax burden × Interest burden × EBIT margin × Total asset turnover × Leverage.*

| Year | Tax burden | Int. burden | EBIT margin | Asset TO | Leverage | ROE |
|---|---|---|---|---|---|---|
| 2014 | 58.96% | 97.49% | 13.98% | 1.44× | 2.14× | 24.72% |
| 2015 | 63.12% | 97.86% | 13.98% | 1.47× | 2.10× | 26.50% |
| 2016 | 52.10% | 97.73% | 11.04% | 1.71× | 2.17× | 20.78% |
| 2017 | 60.50% | 97.49% | 7.56% | 0.99× | 2.15× | 9.53% |

**AMSTERDAM PLC DIAGNOSIS**

2017 ROE collapse (26.5% → 9.5%) came almost entirely from operations: EBIT margin −642 bps AND asset turnover −0.48×. Tax burden and leverage were stable — the story is operational, not financial.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 11, Example 16 (Amsterdam PLC), pp. 436–438.*

SECTION 06

# 3-statement modeling basics

# A 3-statement model is an integrated forecast of the IS, BS, and CFS — built from revenue down and forced to balance every year

*Each statement feeds the next. The cash-flow statement is the integrator — it reconciles the change in cash on the balance sheet back to net income plus working capital.*

1. **Forecast revenue**
   Top-down (market × share) or bottom-up (units × price). Anchor on consensus or management guidance for Yr 1.

2. **Forecast COGS, SG&A, D&A → EBITDA, EBIT**
   Margins usually held stable or mean-reverted to long-run average. D&A linked to gross PP&E schedule.

3. **Compute interest (from debt schedule) → NI**
   Interest = (average debt) × cost of debt. Tax expense = EBT × normalised effective rate.

4. **Build working capital via DSO / DIO / DPO**
   Drives AR, inventory, and AP on balance sheet. Captures cash absorption of revenue growth.

5. **Capex schedule → PP&E → net D&A**
   Capex assumption (% of revenue) maintains asset base. Beginning PP&E + capex − D&A = ending PP&E.

6. **Plug CFS; solve for cash; force BS to tie**
   CFO + CFI + CFF = Δ cash. If BS doesn't balance, revisit the plug in debt or equity until it does.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 12, "Framework for Financial Statement Modeling", pp. 451–465.*

# Working-capital ratios — DSO, DIO, DPO — drive most of the movement between EBITDA and operating cash flow in any forecast

*A fast-growing firm with deteriorating DSO can have positive EBITDA and negative free cash flow simultaneously — the definition of a cash-hungry business.*

$$CCC = DIO + DSO - DPO \quad \text{where} \quad DSO = (AR/Revenue)\times 365$$
$$DIO = (Inv/COGS)\times 365 \quad DPO = (AP/COGS)\times 365$$

*where: Lower CCC = faster cash conversion. A CCC below zero (Amazon, Apple pre-2016) means suppliers fund the working capital.*

| NortCorp forecast | Yr 0 actual | Yr 1 base | Yr 2 base | Yr 3 base | Driver assumption |
|---|---|---|---|---|---|
| Revenue (USD m) | 1,000 | 1,040 | 1,081 | 1,125 | +4% CAGR |
| DSO (days) | 58 | 58 | 58 | 58 | Flat |
| DIO (days) | 72 | 70 | 68 | 66 | −2 days/yr |
| DPO (days) | 45 | 45 | 45 | 45 | Flat |
| CCC (days) | 85 | 83 | 81 | 79 | Improving |
| Net working capital (USD m) | 233 | 237 | 241 | 243 | Tied to CCC |
| Δ NWC absorbed by cash flow | – | +4 | +4 | +2 | Headwind |

# Revenue growth is the single largest lever — +4% base vs +8% bull vs −2% bear deliver wildly different 3-statement outputs

*Scenario analysis is not the same as sensitivity: you move a bundle of coherent assumptions together (growth + margin + capex) to describe a plausible state of the world.*

| NortCorp Yr 3 output | Bear | Base | Bull |
|---|---|---|---|
| Revenue growth assumption | −2% CAGR | +4% CAGR | +8% CAGR |
| EBITDA margin assumption | 16% | 20% | 23% |
| Capex % of revenue | 8% | 6% | 5% |
| Revenue (USD m) | 941 | 1,125 | 1,260 |
| EBITDA (USD m) | 151 | 225 | 290 |
| Net income (USD m) | 62 | 135 | 195 |
| Free cash flow (USD m) | 30 | 130 | 200 |
| Net debt, end of Yr 3 (USD m) | 520 | 420 | 300 |
| Net debt / EBITDA | 3.4× | 1.9× | 1.0× |

**READING**

The three scenarios span a 5× range of free cash flow — and the bear case tips NortCorp above the 3.0× covenant on net debt / EBITDA.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 12, "Scenario Analysis", pp. 478–484.*

KEY TAKEAWAYS

# Deck 8 — FSA III — Taxes, Liabilities, Quality, Ratios & Modeling

- IFRS 16 and ASC 842 capitalise every lease > 12 months as ROU asset = lease liability = PV of payments; only P&L shape differs (IFRS always front-loaded, US GAAP flat for operating).
- Temporary differences generate DTL / DTA; permanent differences only move the effective tax rate; Reston Partners is the canonical accelerated-depreciation DTL worked example.
- Accruals ratio > 10% and CFO/NI < 1.0 are the two simplest earnings-quality red flags — Ventara's 26% / 0.53 combination is a classic pre-restatement pattern.
- Amsterdam PLC 2017: ROE collapse from 26.5% to 9.5% traced to EBIT margin (−642 bps) and asset turnover (−0.48×) — operational, not financial.
- DC plans move pension risk to employees; DB plans keep it with the employer — the PBO rolls forward each year from service + interest + past-service + actuarial + benefits paid.
- Reporting quality (are numbers faithful?) and earnings quality (are economics sustainable?) are independent axes; the 5-tier quality spectrum runs GAAP-faithful → biased → EM → fictitious.
- 4 ratio families — activity, liquidity, solvency, profitability; DuPont decomposes ROE: 3-factor (margin × turnover × leverage) or 5-factor (adds tax burden × interest burden).
- 3-statement model flows revenue → margins → NI → WC → capex → CFS plug; always scenario-analyse revenue growth × margin × capex together, not in isolation.